

Traveling Salesman Problem

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Solution of the practice quiz

We denote N the set of n nodes. The decision variables are binary variables x_{ij} , with $i \in N$, $j \in N$ that are equal to 1 if Anna and Tom visit city j just after city i , and is equal to 0 otherwise. As Anna and Tom want to minimize their travel costs, the objective function is

$$\begin{aligned} \min_{x \in \{0,1\}^{12}} \quad & 150x_{B,Bu} + 170x_{B,P} + 90x_{B,L} \\ & 120x_{L,P} + 200x_{L,Bu} + 90x_{L,B} \\ & 70x_{Bu,P} + 150x_{Bu,B} + 200x_{Bu,L} \\ & 170x_{P,B} + 120x_{P,L} + 70x_{Bu,P}. \end{aligned}$$

Each city must have exactly one successor, that is

$$\begin{aligned} x_{B,Bu} + x_{B,P} + x_{B,L} &= 1, \\ x_{L,P} + x_{L,Bu} + x_{L,B} &= 1, \\ x_{Bu,P} + x_{Bu,B} + x_{Bu,L} &= 1, \\ x_{P,B} + x_{P,L} + x_{Bu,P} &= 1. \end{aligned}$$

Furthermore, each city must have exactly one predecessor, that is

$$\begin{aligned} x_{Bu,B} + x_{P,B} + x_{L,B} &= 1, \\ x_{P,L} + x_{Bu,L} + x_{B,L} &= 1, \\ x_{P,Bu} + x_{B,Bu} + x_{L,Bu} &= 1, \\ x_{B,P} + x_{L,P} + x_{Bu,P} &= 1. \end{aligned}$$

Finally, to eliminate subtours, we introduce the integer variables y_i , for each city i except London, modeling the position of city i in the tour. The

subtour elimination constraint imposes that, if $x_{ij} = 1$, then $y_j \geq y_i + 1$. It can be modeled as

$$x_{ij}(n - 1) \leq y_j - y_i + n - 2, \quad \forall i, j \in N \setminus \{\text{London}\},$$

where $n = 4$. If $x_{ij} = 1$, that is, if city j is visited just after city i , the constraint becomes

$$1 \cdot 3 \leq y_j - y_i + 4 - 2,$$

or, equivalently,

$$y_j \geq y_i + 1,$$

that is, the position of city j in the tour must be after city i .

If $x_{ij} = 0$, that is if city j is not the city visited after city i , the constraint becomes

$$0 \leq y_j - y_i + 4 - 2,$$

or, equivalently,

$$y_i - y_j \leq 2,$$

which is always verified. Indeed, as London is not numbered, the numbering will be 1,2,3, and the largest difference is 2. In general, any difference $y_j - y_i$ is less or equal to $n - 2$.

The subtour elimination constraints are:

$$3x_{B,Bu} + y_B - y_{Bu} \leq 2,$$

$$3x_{Bu,B} + y_{Bu} - y_B \leq 2,$$

$$3x_{P,B} + y_P - y_B \leq 2,$$

$$3x_{B,P} + y_B - y_P \leq 2,$$

$$3x_{Bu,P} + y_{Bu} - y_P \leq 2,$$

$$3x_{P,Bu} + y_P - y_{Bu} \leq 2.$$

To sum up, the mathematical model for the given problem is the following:

$$\begin{aligned}
& \min_{x \in \{0,1\}^{12}} 150x_{B,Bu} + 170x_{B,P} + 90x_{B,L} \\
& \quad 120x_{L,P} + 200x_{L,Bu} + 90x_{L,B} \\
& \quad 70x_{Bu,P} + 150x_{Bu,B} + 200x_{Bu,L} \\
& \quad 170x_{P,B} + 120x_{P,L} + 70x_{Bu,P} \\
& \text{subject to} \\
& \quad x_{B,Bu} + x_{B,P} + x_{B,L} = 1, \\
& \quad x_{L,P} + x_{L,Bu} + x_{L,B} = 1, \\
& \quad x_{Bu,P} + x_{Bu,B} + x_{Bu,L} = 1, \\
& \quad x_{P,B} + x_{P,L} + x_{Bu,P} = 1, \\
& \quad x_{Bu,B} + x_{P,B} + x_{L,B} = 1, \\
& \quad x_{P,L} + x_{Bu,L} + x_{B,L} = 1, \\
& \quad x_{P,Bu} + x_{B,Bu} + x_{L,Bu} = 1, \\
& \quad x_{B,P} + x_{L,P} + x_{Bu,P} = 1, \\
& \quad 3x_{B,Bu} + y_B - y_{Bu} \leq 2, \\
& \quad 3x_{Bu,B} + y_{Bu} - y_B \leq 2, \\
& \quad 3x_{P,B} + y_P - y_B \leq 2, \\
& \quad 3x_{B,P} + y_B - y_P \leq 2, \\
& \quad 3x_{Bu,P} + y_{Bu} - y_P \leq 2, \\
& \quad 3x_{P,Bu} + y_P - y_{Bu} \leq 2, \\
& \quad y_i \in \{1, 2, 3\}, \quad \forall i \in N \setminus \{L\}, \\
& \quad x_{i,j} \in \{0, 1\}, \quad \forall i, j \in N.
\end{aligned}$$

Consider the tour London $\rightarrow$ Barcelona $\rightarrow$ Budapest $\rightarrow$ Prague. The corresponding decision variables are

$$\begin{array}{lll}
x_{L,B} = 1, & x_{L,Bu} = 0, & x_{L,P} = 0, \\
x_{B,L} = 0, & x_{B,Bu} = 1, & x_{B,P} = 0, \\
x_{Bu,L} = 0, & x_{Bu,B} = 0, & x_{Bu,P} = 1, \\
x_{P,L} = 1, & x_{P,B} = 0, & x_{P,Bu} = 0, \\
y_B = 1, & y_{Bu} = 2, & y_P = 3.
\end{array}$$

We can verify the constraints:

$$\begin{aligned}
x_{B,Bu} + x_{B,P} + x_{B,L} &= 1 + 0 + 0 &= 1, \\
x_{L,P} + x_{L,Bu} + x_{L,B} &= 0 + 0 + 1 &= 1, \\
x_{Bu,P} + x_{Bu,B} + x_{Bu,L} &= 1 + 0 + 0 &= 1, \\
x_{P,B} + x_{P,L} + x_{Bu,P} &= 0 + 1 + 0 &= 1, \\
x_{Bu,B} + x_{P,B} + x_{L,B} &= 0 + 0 + 1 &= 1, \\
x_{P,L} + x_{Bu,L} + x_{B,L} &= 1 + 0 + 0 &= 1, \\
x_{P,Bu} + x_{B,Bu} + x_{L,Bu} &= 0 + 1 + 0 &= 1, \\
x_{B,P} + x_{L,P} + x_{Bu,P} &= 0 + 0 + 1 &= 1, \\
3x_{B,Bu} + y_B - y_{Bu} &= 3 + 1 - 2 = 2 &\leq 2, \\
3x_{Bu,B} + y_{Bu} - y_B &= 2 - 1 = 1 &\leq 2, \\
3x_{P,B} + y_P - y_B &= 3 - 1 = 2 &\leq 2, \\
3x_{B,P} + y_B - y_P &= 1 - 3 = -2 &\leq 2, \\
3x_{Bu,P} + y_{Bu} - y_P &= 3 + 2 - 3 = 2 &\leq 2, \\
3x_{P,Bu} + y_P - y_{Bu} &= 3 - 2 = 1 &\leq 2.
\end{aligned}$$